Supplementary Section S3: Convergence and Centering Analysis of EMOJI

S3 Convergence analysis of EMOJI

Scope. This section analyzes the core EMOJI recursion in the manuscript. It has two purposes: (i) to establish a finite-time attenuation bound for the centered second-moment state relative to an uncentered reference computed from the same prediction-error sequence, and (ii) to derive deterministic and stochastic stationarity guarantees under explicit assumptions. The attenuation comparison is a same-sequence statement; it does not assert that two independently trained runs generate identical prediction errors. The convergence results apply to the core adaptive update and do not include the separate decoupled weight-decay step used in the experiments or any gradient-clipping operation that may be applied during training.

S3.1 Algorithm and notation

We study the elementwise core EMOJI recursion with zero initialization. For a parameter vector $\theta_t \in \mathbb{R}^d$, let g_t denote the gradient supplied to the optimizer at iteration t. For a fixed centering coefficient $\alpha \in [0,1]$, write

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \qquad d_t = g_t - m_t, \tag{S3.1}$$

$$v_t = \beta_1 v_{t-1} + (1-\beta_1) d_t, \qquad q_t = d_t - \alpha v_t, \tag{S3.2}$$

$$j_t = \beta_2 j_{t-1} + (1-\beta_2) q_t^{\odot 2}, \qquad \widehat{m}_t = \frac{m_t}{1-\beta_1^t}, \qquad \widehat{j}_t = \frac{j_t}{1-\beta_2^t}, \tag{S3.3}$$

$$\theta_t = \theta_{t-1} - \eta_t A_t \widehat{m}_t, \qquad A_t = \operatorname{diag}\left(\frac{1}{\sqrt{\widehat{j}_t} + \epsilon \mathbf{1}} \right). \tag{S3.4}$$

All squares, square roots, and divisions in these equations are coordinatewise. The proposed EMOJI algorithm in the manuscript corresponds to $\alpha = 1$; $\alpha = 0$ is the uncentered ablation. We assume $\beta_1, \beta_2 \in [0,1)$, $\alpha \in [0,1]$, and $\epsilon > 0$. The stochastic-gradient vector g_t is the quantity supplied to the optimizer; it is the exact gradient in the deterministic analysis and a stochastic oracle output in the stochastic analysis.

The bias-corrected first moment has the exact representation

$$\widehat{m}_t = \sum_{s=1}^{t} w_{t,s} g_s, \qquad w_{t,s} = \frac{(1-\beta_1)\beta_1^{t-s}}{1-\beta_1^t}, \qquad \sum_{s=1}^{t} w_{t,s} = 1. \tag{S3.5}$$

Thus $\widehat{m}_t$ is a convex combination of the current and past supplied gradients. A useful exact identity, which follows from the fact that m_t includes the current gradient, is

$$d_t = \beta_1 (g_t - m_{t-1}), \qquad q_t = \big[1 - \alpha(1-\beta_1)\big] d_t - \alpha \beta_1 v_{t-1}. \tag{S3.6}$$

In particular, for the proposed setting $\alpha = 1$,

$$q_t = d_t - v_t = \beta_1(d_t - v_{t-1}). \tag{S3.7}$$

This identity is important: the quantity accumulated by j_t is a filtered residual, not in general an unbiased estimate of the central variance of d_t.

S3.2 Finite-time attenuation induced by centering

The centering mechanism can be compared with an uncentered second-moment accumulator driven by the *same* prediction-error sequence. For this purpose, define

$$j_t^{\mathrm{raw}} = \beta_2 j_{t-1}^{\mathrm{raw}} + (1-\beta_2) d_t^{\odot 2}, \qquad j_0^{\mathrm{raw}} = 0,$$

while j_t^{cen} denotes the EMOJI state obtained from $q_t = d_t - v_t$ with $v_0 = j_0^{\mathrm{cen}} = 0$. Let the corresponding bias-corrected states be denoted by $\widehat{j}_t^{\mathrm{raw}}$ and $\widehat{j}_t^{\mathrm{cen}}$.

Proposition S3.1 (Finite-time attenuation of the centered second moment)**.** *Let $\{d_t\}_{t=1}^T$ be any real-valued scalar sequence, let $v_t = \beta_1 v_{t-1} + (1-\beta_1) d_t$ with $v_0 = 0$, and let $q_t = d_t - v_t$. Suppose*

$$0 < \beta_1 \le \beta_2 < 1.$$

Define

$$r := \frac{\beta_1}{\sqrt{\beta_2}}, \qquad \gamma := \frac{\beta_1 + r}{1 + r}.$$

Then $0 < r < 1$, $0 < \gamma < 1$, and for every $t \le T$,

$$\widehat{j}_t^{\mathrm{cen}} \le \gamma^2 \widehat{j}_t^{\mathrm{raw}}. \tag{S3.8}$$

For vector-valued prediction errors, the inequality holds coordinatewise. No stationarity, independence, or distributional assumption on d_t is required.

Proof. The EMA recursion gives the exact finite-history representation

$$q_s = \beta_1 d_s - (1-\beta_1) \sum_{\ell=1}^{s-1} \beta_1^{\ell} d_{s-\ell}.$$

Fix a terminal time t and define the exponentially weighted input and output sequences

$$x_s = \beta_2^{(t-s)/2} d_s, \qquad y_s = \beta_2^{(t-s)/2} q_s, \qquad 1 \le s \le t.$$

With $r = \beta_1/\sqrt{\beta_2}$, the relation between them is the first t samples of the causal linear filter

$$y_s = \beta_1 x_s - (1-\beta_1) \sum_{\ell=1}^{s-1} r^{\ell} x_{s-\ell}.$$

Its transfer function is

$$H(z) = \beta_1 - (1-\beta_1) \sum_{\ell=1}^{\infty} r^{\ell} z^{-\ell} = \frac{\beta_1 - r z^{-1}}{1 - r z^{-1}}.$$

The series is absolutely summable because $r < 1$. On the unit circle, writing $c = \cos\omega$,

$$|H(e^{i\omega})|^2 = \frac{\beta_1^2 + r^2 - 2\beta_1 rc}{1 + r^2 - 2rc}.$$

Differentiating this expression with respect to c gives

$$\frac{2r(\beta_1 - 1)(\beta_1 - r^2)}{(1 + r^2 - 2rc)^2} \leq 0,$$

where the inequality follows from $r^2 = \beta_1^2/\beta_2 \leq \beta_1$. Hence the maximum occurs at $c = -1$, and

$$\sup_{\omega} |H(e^{i\omega})| = \frac{\beta_1 + r}{1 + r} = \gamma.$$

Extend x_s by zero for $s > t$ and let the filter generate its full causal output. By Parseval's identity and the induced ℓ_2 bound for a stable linear filter,

$$\sum_{s=1}^{t} y_s^2 \leq \sum_{s=1}^{\infty} y_s^2 \leq \gamma^2 \sum_{s=1}^{\infty} x_s^2 = \gamma^2 \sum_{s=1}^{t} x_s^2.$$

Substituting the definitions of x_s and y_s yields

$$\sum_{s=1}^{t} \beta_2^{t-s} q_s^2 \leq \gamma^2 \sum_{s=1}^{t} \beta_2^{t-s} d_s^2.$$

Multiplication by the common positive bias-correction factor $(1 - \beta_2)/(1 - \beta_2^t)$ proves (S3.8). The vector result follows by applying the scalar argument independently to each coordinate. □

Corollary S3.2 (Denominator reduction under a common prediction-error history)**.** *Under the assumptions of Proposition S3.1, define*

$$D_{t,i}^{\mathrm{cen}} = \sqrt{\widehat{j}_{t,i}^{\,\mathrm{cen}}} + \epsilon, \qquad D_{t,i}^{\mathrm{raw}} = \sqrt{\widehat{j}_{t,i}^{\,\mathrm{raw}}} + \epsilon.$$

Then

$$D_{t,i}^{\mathrm{cen}} \leq \gamma \sqrt{\widehat{j}_{t,i}^{\,\mathrm{raw}}} + \epsilon \leq D_{t,i}^{\mathrm{raw}}. \tag{S3.9}$$

The second inequality is strict whenever $\widehat{j}_{t,i}^{\,\mathrm{raw}} > 0$. Thus the centered adaptive multiplier is at least as large as its uncentered same-history counterpart. Equivalently, when the raw state is nonzero,

$$\frac{D_{t,i}^{\mathrm{cen}}}{D_{t,i}^{\mathrm{raw}}} \leq \frac{\gamma \sqrt{\widehat{j}_{t,i}^{\,\mathrm{raw}}} + \epsilon}{\sqrt{\widehat{j}_{t,i}^{\,\mathrm{raw}}} + \epsilon} < 1.$$

The additive ϵ is not multiplied by γ; consequently, the proposition does not imply that the complete denominator is reduced by the factor γ. For example, when $\beta_1 = 0.9$ and $\beta_2 = 0.98$, $\gamma \approx 0.9476$ and $\gamma^2 \approx 0.8980$, so the centered second-moment estimate is bounded by approximately 89.8% of the raw estimate for the same prediction-error history.

Remark S3.3 (Scope of the attenuation comparison)**.** *The bound concerns the accumulated, exponentially weighted second moment, not each instantaneous residual: it is possible to have $q_t^2 > d_t^2$ at an individual iteration. The proposition controls the cumulative state at every finite time. It compares centered and raw accumulators on a common d_t sequence; it does not guarantee the same inequality between two independent training runs after their parameter trajectories have diverged.*

S3.3 Standing assumptions and uniform bounds

The following assumptions are sufficient for the deterministic results.

(A1) $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable and L-smooth along every update segment $[\theta_{t-1}, \theta_t]$; that is, $\|\nabla f(x) - \nabla f(y)\| \le L \|x - y\|$ for all x, y on each such segment.

(A2) f is bounded below on the iterates: $f(\theta) \ge f_{\inf} > -\infty$.

(A3) The gradients along the trajectory are bounded: $\|\nabla f(\theta_{t-1})\|_2 \le G$ for all t under consideration.

(A4) The algorithm uses the exact recursion (S3.1)–(S3.4), zero initial moments, fixed $\beta_1, \beta_2 \in [0, 1)$, $\alpha \in [0, 1]$, a strictly positive ϵ, and positive learning rates $\eta_t > 0$.

The bounded-gradient assumption can be replaced by boundedness of the iterates together with continuity of the gradient on a compact set. It is a trajectory assumption, not a claim that arbitrary neural-network objectives have globally bounded gradients.

Lemma S3.4 (Uniform bounds on EMOJI states and preconditioner)**.** *Suppose $\|g_t\|_2 \le G$ at every iteration and that the algorithm satisfies (A4). Define the lower spectral constant $\underline{\mu}$ by*

$$\underline{\mu} := \begin{cases} (2\gamma G + \epsilon)^{-1}, & \alpha = 1 \text{ and } 0 < \beta_1 \le \beta_2 < 1, \\ \epsilon^{-1}, & \alpha = 1 \text{ and } \beta_1 = 0, \\ [2G(1+\alpha) + \epsilon]^{-1}, & \text{otherwise}, \end{cases} \qquad M := \epsilon^{-1}, \qquad P := G\epsilon^{-1}, \tag{S3.10}$$

where γ is defined in Proposition S3.1 whenever it is used. Then, for every t,

$$\underline{\mu} I \preceq A_t \preceq M I, \qquad \|\widehat{m}_t\|_2 \le G, \qquad \|A_t \widehat{m}_t\|_2 \le P. \tag{S3.11}$$

In particular, for the proposed centered setting with $\beta_2 \ge \beta_1 > 0$, the attenuation result gives the sharper lower bound $\underline{\mu} = (2\gamma G + \epsilon)^{-1}$.

Proof. The recursion for m_t is an EMA with nonnegative coefficients whose sum is $1 - \beta_1^t \le 1$. Hence $\|m_t\|_2 \le G$, and therefore $\|d_t\|_2 = \|g_t - m_t\|_2 \le 2G$. The same argument applied to the EMA for v_t gives $\|v_t\|_2 \le 2G$. Thus, for general α, $\|q_t\|_2 \le 2G(1+\alpha)$, which yields $\sqrt{\widehat{j}_{t,i}} \le 2G(1+\alpha)$.

For $\alpha = 1$ and $0 < \beta_1 \le \beta_2$, Proposition S3.1 applies to the actual prediction-error history. The uncentered reference satisfies $\widehat{j}^{\,\mathrm{raw}}_{t,i} \le 4G^2$, because it is a bias-corrected EMA of $d_{t,i}^2$ and $|d_{t,i}| \le 2G$. Therefore

$$\widehat{j}^{\,\mathrm{cen}}_{t,i} \le \gamma^2 \widehat{j}^{\,\mathrm{raw}}_{t,i} \le 4\gamma^2 G^2,$$

so $\sqrt{\widehat{j}_{t,i}} \le 2\gamma G$. If $\alpha = 1$ and $\beta_1 = 0$, then $m_t = g_t$, $d_t = v_t = j_t = 0$, and the denominator equals ϵ.

In all cases, the denominator is at least ϵ and at most $\underline{\mu}^{-1}$. Taking coordinatewise reciprocals proves the spectral bounds. Finally, the bias-corrected first moment is a convex combination of vectors with norm at most G, so $\|\widehat{m}_t\|_2 \le G$. The update-direction bound follows from $\|A_t\|_2 \le M$. □

Lemma S3.5 (Momentum-lag bound)**.** *Suppose that f is L-smooth and that the deterministic gradients satisfy (A3). Let*

$$e_t := \widehat{m}_t - \nabla f(\theta_{t-1}), \qquad \Delta_t := \frac{LG}{\epsilon} \sum_{s=1}^{t} w_{t,s} \sum_{k=s}^{t-1} \eta_k, \tag{S3.12}$$

where an empty inner sum is zero. Then $\|e_t\|_2 \le \Delta_t$. If $\eta_t \equiv \eta$, then

$$\Delta_t \le K\eta, \qquad K := \frac{LG}{\epsilon} \frac{\beta_1}{1 - \beta_1}. \tag{S3.13}$$

Proof. Using (S3.5) and writing $h_t = \nabla f(\theta_{t-1})$,

$$e_t = \sum_{s=1}^{t} w_{t,s}(h_s - h_t).$$

By smoothness and the update bound in Lemma S3.4,

$$\|h_s - h_t\|_2 \le \sum_{k=s}^{t-1} \|h_k - h_{k+1}\|_2 \le L \sum_{k=s}^{t-1} \|\theta_k - \theta_{k-1}\|_2 \le L \sum_{k=s}^{t-1} \eta_k \|A_k \widehat{m}_k\|_2 \le \frac{LG}{\epsilon} \sum_{k=s}^{t-1} \eta_k.$$

Averaging with the nonnegative weights gives (S3.12). For constant η, the coefficient of η is the expected age of the geometric weights:

$$\sum_{s=1}^{t} w_{t,s}(t-s) = \frac{\sum_{r=0}^{t-1} r(1-\beta_1)\beta_1^r}{1-\beta_1^t} \le \frac{\beta_1}{1-\beta_1}.$$

This proves (S3.13). □

S3.4 Deterministic nonconvex convergence

Theorem S3.6 (Finite-time stationarity bound for deterministic EMOJI)**.** *Suppose (A1)–(A4) hold and $g_t = \nabla f(\theta_{t-1})$. Let $S_1 = \sum_{t=1}^{T} \eta_t$ and $S_2 = \sum_{t=1}^{T} \eta_t^2$. Then*

$$\frac{\sum_{t=1}^{T} \eta_t \|\nabla f(\theta_{t-1})\|_2^2}{S_1} \le \frac{2\big(f(\theta_0) - f_{\inf}\big)}{\underline{\mu} S_1} + \frac{M^2}{\underline{\mu}^2} \frac{\sum_{t=1}^{T} \eta_t \Delta_t^2}{S_1} + \frac{LP^2}{\underline{\mu}} \frac{S_2}{S_1}, \tag{S3.14}$$

where Δ_t is defined in (S3.12)*. In particular, for a constant learning rate $\eta_t = \eta$,*

$$\min_{1 \le t \le T} \|\nabla f(\theta_{t-1})\|_2^2 \le \frac{2\big(f(\theta_0) - f_{\inf}\big)}{\underline{\mu} \eta T} + \frac{M^2}{\underline{\mu}^2} K^2 \eta^2 + \frac{LP^2}{\underline{\mu}} \eta. \tag{S3.15}$$

For fixed $\epsilon > 0$, $\beta_1 < 1$, G, and L, choosing $\eta = \Theta(T^{-1/2})$ yields an $\mathcal{O}(T^{-1/2})$ bound on the minimum squared gradient norm.

Proof. Let $h_t = \nabla f(\theta_{t-1})$ and $e_t = \widehat{m}_t - h_t$. By Lemma S3.4,

$$\langle h_t, A_t \widehat{m}_t \rangle = \langle h_t, A_t h_t \rangle + \langle h_t, A_t e_t \rangle \ge \underline{\mu} \|h_t\|_2^2 - M \|h_t\|_2 \|e_t\|_2 .$$

Young's inequality gives

$$\langle h_t, A_t \widehat{m}_t \rangle \ge \frac{\underline{\mu}}{2} \|h_t\|_2^2 - \frac{M^2}{2\underline{\mu}} \|e_t\|_2^2 . \tag{S3.16}$$

The L-smoothness inequality and the update equation imply

$$f(\theta_t) \le f(\theta_{t-1}) - \eta_t \langle h_t, A_t \widehat{m}_t \rangle + \frac{L\eta_t^2}{2} \|A_t \widehat{m}_t\|_2^2 .$$

Combining this inequality with (S3.16), Lemmas S3.4–S3.5, and $f(\theta_T) \ge f_{\inf}$ yields

$$\frac{\underline{\mu}}{2} \sum_{t=1}^{T} \eta_t \|h_t\|_2^2 \le f(\theta_0) - f_{\inf} + \frac{M^2}{2\underline{\mu}} \sum_{t=1}^{T} \eta_t \Delta_t^2 + \frac{LP^2}{2} \sum_{t=1}^{T} \eta_t^2.$$

Multiplication by $2/(\underline{\mu} S_1)$ proves (S3.14). The constant-step result follows from $\Delta_t \le K\eta$, $S_1 = T\eta$, and $S_2 = T\eta^2$. □

Remark S3.7 (Role of the centering attenuation bound)**.** *The descent proof uses the second-moment state to establish uniform spectral bounds on A_t. For the proposed setting $\alpha = 1$ with $\beta_2 \geq \beta_1$, Proposition S3.1 sharpens the lower spectral bound through $\underline{\mu} = (2\gamma G + \epsilon)^{-1}$. This improves the corresponding sufficient bound on the preconditioner relative to the crude estimate obtained from $|d_t - v_t| \leq 4G$. It does not establish that two independently evolving optimizer trajectories have a fixed ordering of their updates, nor does it prove a smaller empirical loss or a universally faster convergence rate. The centering result is a same-history denominator comparison; the stationarity theorem is a separate guarantee for the actual core update.*

Corollary S3.8 (Diminishing-step convergence to first-order stationarity)**.** *Under the assumptions of Theorem S3.6, suppose $\eta_t = \eta_0(t+1)^{-p}$ for some $p \in (1/2, 1]$. Then*

$$\sum_{t=1}^{\infty} \eta_t \left\|\nabla f(\theta_{t-1})\right\|_2^2 < \infty, \qquad \lim_{t\to\infty} \left\|\nabla f(\theta_{t-1})\right\|_2 = 0.$$

Moreover, the weighted-average bound implies

$$\min_{1\leq t\leq T} \left\|\nabla f(\theta_{t-1})\right\|_2^2 = \begin{cases} \mathcal{O}(T^{p-1}), & p \in (1/2, 1), \\ \mathcal{O}(1/\log T), & p = 1. \end{cases}$$

Proof. For fixed $\beta_1 < 1$, the geometrically weighted lag in (S3.12) satisfies $\Delta_t = \mathcal{O}(\eta_t)$ for a polynomially decreasing schedule with $p \in (0, 1]$. To see this, split the lag index $r = t - s$ into $r \leq t/2$ and $r > t/2$. In the first range, $\eta_{t-r} \leq 2^p \eta_t$ and the geometric first moment $\sum_{r\geq 0} r\beta_1^r$ is finite. In the second range, the geometric tail is exponentially small and dominates the at-most-polynomial growth of $\sum_{k=1}^{t} \eta_k$. Thus $\sum_t \eta_t \Delta_t^2 = \mathcal{O}(\sum_t \eta_t^3) < \infty$. Also $\sum_t \eta_t^2 < \infty$ and $\sum_t \eta_t = \infty$. The telescoping inequality in the proof of Theorem S3.6 therefore yields $\sum_t \eta_t \left\|h_t\right\|_2^2 < \infty$.

By Lemma S3.4, $\left\|\theta_t - \theta_{t-1}\right\|_2 \leq P\eta_t$, so smoothness gives $|\left\|h_{t+1}\right\|_2 - \left\|h_t\right\|_2| \leq LP\eta_t$. If $\left\|h_t\right\|_2$ did not tend to zero, this bounded variation per unit step length, together with $\sum_t \eta_t = \infty$, would force infinitely many disjoint intervals on which $\left\|h_t\right\|_2$ stays bounded away from zero; each such interval would contribute a fixed positive amount to $\sum_t \eta_t \left\|h_t\right\|_2^2$, a contradiction. Hence $\left\|h_t\right\|_2 \to 0$. □

Corollary S3.9 (Polyak–Lojasiewicz objectives)**.** *In addition to the assumptions of Theorem S3.6, suppose that f satisfies the PL inequality*

$$\left\|\nabla f(\theta)\right\|_2^2 \geq 2\mu_{\mathrm{PL}}\big(f(\theta) - f_*\big)$$

on the trajectory, where $f_* := \inf_\theta f(\theta)$ is finite. Then, whenever $\mu_{\mathrm{PL}}\underline{\mu}\eta_t \leq 1$,

$$f(\theta_t) - f_* \leq (1 - \mu_{\mathrm{PL}}\underline{\mu}\eta_t)\big(f(\theta_{t-1}) - f_*\big) + \frac{M^2}{2\underline{\mu}}\eta_t\Delta_t^2 + \frac{LP^2}{2}\eta_t^2. \tag{S3.17}$$

Consequently, for $\eta_t = \eta_0(t+1)^{-p}$ with $p \in (1/2, 1]$, the objective gap converges to zero. For a sufficiently small constant η, the objective gap contracts geometrically to an $\mathcal{O}(\eta)$ neighborhood, as follows from the fixed point of (S3.17)*.*

Proof. Apply the one-step descent inequality established in the proof of Theorem S3.6 and then use the PL inequality. The diminishing-step conclusion follows because $\sum_t \eta_t = \infty$ while the additive terms in (S3.17) are summable by Corollary S3.8. For constant η, Lemma S3.5 bounds the additive term by $\mathcal{O}(\eta^2)$ per step, while the contraction coefficient differs from one by $\mu_{\mathrm{PL}}\underline{\mu}\eta$; their ratio is $\mathcal{O}(\eta)$. □

Remark S3.10 (The $\beta_1 = 0$ limiting case)**.** *If $\beta_1 = 0$, then $m_t = g_t$, $d_t = v_t = j_t = 0$, and EMOJI reduces exactly to gradient descent with effective learning rate η_t/ϵ. Thus the centering and second-moment mechanisms are inactive in this limit. This is a useful sanity check, but it is not representative of the intended momentum setting.*

S3.5 Stochastic convergence and the noise floor

Let $f(\theta) = \mathbb{E}_\zeta[\ell(\theta;\zeta)]$ be the population objective and let

$$g_t = \nabla f(\theta_{t-1}) + \xi_t, \qquad \mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0, \qquad \mathbb{E}[\|\xi_t\|_2^2 \mid \mathcal{F}_{t-1}] \le \sigma_t^2,$$

where $\{\xi_t\}$ is a martingale-difference sequence. Assume, in addition, that $\|g_t\|_2 \le G$ almost surely and that f is L-smooth on a region containing the iterates and update segments, and is bounded below on the iterates. Define $e_t = \widehat{m}_t - \nabla f(\theta_{t-1})$ and Δ_t as in (S3.12).

Lemma S3.11 (Mean-square tracking error)**.** *Under the preceding stochastic assumptions,*

$$\mathbb{E}\,\|e_t\|_2^2 \le 2\Delta_t^2 + 2\sum_{s=1}^{t} w_{t,s}^2 \sigma_s^2. \tag{S3.18}$$

Moreover, if $\sigma_s^2 \le \sigma^2/B_s$ (e.g. a mini-batch of size B_s), then the second term is at most $2\sigma^2 \sum_s w_{t,s}^2/B_s$. For constant $B_s = B$,

$$W_t := \sum_{s=1}^{t} w_{t,s}^2 = \frac{1-\beta_1}{1+\beta_1}\frac{1+\beta_1^t}{1-\beta_1^t} \longrightarrow \frac{1-\beta_1}{1+\beta_1}. \tag{S3.19}$$

Proof. Using (S3.5), decompose

$$e_t = \underbrace{\sum_{s=1}^{t} w_{t,s}\big(\nabla f(\theta_{s-1}) - \nabla f(\theta_{t-1})\big)}_{b_t} + \underbrace{\sum_{s=1}^{t} w_{t,s}\xi_s}_{n_t}.$$

The same argument as in Lemma S3.5 gives $\|b_t\|_2 \le \Delta_t$. Since the weights are deterministic and the noise is a martingale difference, the cross terms vanish in expectation and $\mathbb{E}\,\|n_t\|_2^2 \le \sum_s w_{t,s}^2\sigma_s^2$. Finally, use $\|b_t + n_t\|^2 \le 2\,\|b_t\|^2 + 2\,\|n_t\|^2$. The formula for W_t follows by summing the geometric series. □

Theorem S3.12 (Stochastic EMOJI: finite-time stationarity bound)**.** *Under the stochastic assumptions above, let $S_1 = \sum_{t=1}^T \eta_t$ and $S_2 = \sum_{t=1}^T \eta_t^2$. Then*

$$\frac{\sum_{t=1}^T \eta_t \mathbb{E}\,\|\nabla f(\theta_{t-1})\|_2^2}{S_1} \le \frac{2(f(\theta_0) - f_{\inf})}{\underline{\mu} S_1} + 2\frac{M^2}{\underline{\mu}^2}\frac{\sum_{t=1}^T \eta_t\left(\Delta_t^2 + \sum_{s=1}^t w_{t,s}^2\sigma_s^2\right)}{S_1} + \frac{LP^2}{\underline{\mu}}\frac{S_2}{S_1}. \tag{S3.20}$$

For a constant learning rate η and constant mini-batch size B, this gives

$$\frac{1}{T}\sum_{t=1}^T \mathbb{E}\,\|\nabla f(\theta_{t-1})\|_2^2 \le \frac{2(f(\theta_0) - f_{\inf})}{\underline{\mu}\eta T} + 2\frac{M^2}{\underline{\mu}^2}\left(K^2\eta^2 + \frac{\sigma^2}{B}\frac{1}{T}\sum_{t=1}^T W_t\right) + \frac{LP^2}{\underline{\mu}}\eta. \tag{S3.21}$$

Thus, for fixed B and fixed η, the limiting upper bound on the time-averaged expected squared gradient norm contains a step-size- and variance-dependent contribution proportional to $\sigma^2 B^{-1}(1-\beta_1)/(1+\beta_1)$. This is an average-stationarity bound; it does not imply pointwise convergence of the iterates or of every gradient norm. Increasing batch sizes or variance reduction can remove the noise contribution when they make the weighted noise term vanish.

Proof. The uniform matrix bounds in Lemma S3.4 hold almost surely. Therefore the alignment inequality (S3.16) is pathwise valid even though A_t is random and depends on the current stochastic gradient. Apply the smoothness inequality pathwise, take expectations, use $\mathbb{E}f(\theta_T) \geq f_{\inf}$, and then apply Lemma S3.11. This yields (S3.20). For constant η, Lemma S3.5 gives $\Delta_t \leq K\eta$, and (S3.19) gives (S3.21). □

Remark S3.13 (What is and is not proved in the stochastic setting)**.** *The theorem is a finite-time bound under a bounded stochastic-gradient oracle. With a fixed nonzero variance and fixed batch size, the limiting time-averaged bound retains a variance-dependent contribution; it does not establish convergence of the iterates or pointwise convergence to an exact stationary point. This is not merely a technicality: for $\beta_1 > 0$, the current preconditioner A_t is constructed from a residual that uses the same stochastic sample as the update, so it is generally correlated with the noise. Unbiasedness of g_t alone does not make $A_t\widehat{m}_t$ an unbiased preconditioned gradient. The separate expected objective-gap result in Theorem S3.14 requires the additional relative-tracking-error condition stated there. Neither that condition nor exact stochastic convergence follows from unbiasedness alone.*

Theorem S3.14 (Stochastic PL convergence under relative tracking error)**.** *Suppose the assumptions of Theorem S3.12 hold and f satisfies the PL inequality with constant $\mu_{\mathrm{PL}} > 0$. Assume additionally that, for some $\rho \geq 0$ and deterministic $\nu_t \geq 0$,*

$$\mathbb{E}\,\|e_t\|_2^2 \leq \rho\,\mathbb{E}\,\|\nabla f(\theta_{t-1})\|_2^2 + \nu_t, \qquad \rho < \Big(\frac{\underline{\mu}}{M}\Big)^2. \tag{S3.22}$$

Let $\kappa = 1 - (M/\underline{\mu})^2\rho > 0$. If $\mu_{\mathrm{PL}}\underline{\mu}\kappa\eta_t \leq 1$, $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$, and $\sum_t \eta_t\nu_t < \infty$, then

$$\mathbb{E}\big[f(\theta_t) - f_*\big] \longrightarrow 0.$$

Proof. Repeat the pathwise descent proof but use (S3.22) after taking expectations. The coefficient of $\mathbb{E}\,\|\nabla f(\theta_{t-1})\|^2$ becomes $\underline{\mu}/2 - M^2\rho/(2\underline{\mu}) = \underline{\mu}\kappa/2$. The PL inequality therefore gives

$$\mathbb{E}[F_t] \leq (1 - \mu_{\mathrm{PL}}\underline{\mu}\kappa\eta_t)\mathbb{E}[F_{t-1}] + \frac{M^2}{2\underline{\mu}}\eta_t\nu_t + \frac{LP^2}{2}\eta_t^2, \qquad F_t = f(\theta_t) - f_*.$$

The contraction coefficients have a divergent sum, while the additive terms are summable. The standard deterministic comparison argument for nonnegative sequences then implies $\mathbb{E}[F_t] \to 0$. □

S3.6 Statistical interpretation of the centered residual

The attenuation result above does not make j_t an unbiased estimator of the central variance. Since v_t includes the current d_t and uses the same decay parameter as m_t,

$$d_t - v_t = \beta_1(d_t - v_{t-1}).$$

Thus the residual is a temporally filtered (high-pass) version of d_t. Its second moment depends on the temporal autocovariance of d_t. The finite-time attenuation proposition is a deterministic energy comparison and does not require this stationary diagnostic model. In addition, if $\mathbb{E}[d_t] = \mu_d$ is constant and $v_0 = 0$, then

$$\mathbb{E}[v_t] = (1 - \beta_1^t)\mu_d, \qquad \mathbb{E}[d_t - v_t] = \beta_1^t\mu_d,$$

so the uncorrected v_t has a startup bias.

Proposition S3.15 (Stationary white-noise calculation for the actual EMOJI residual)**.** *Consider a diagnostic stationary regime in which $g_t = \mu + \xi_t$, where $\{\xi_t\}$ are i.i.d., zero-mean, and have coordinatewise variance σ^2. Let $a = \beta_1 \in (0,1)$ and assume the EMA states have reached stationarity. For the exact EMOJI definitions $d_t = g_t - m_t$ and $q_t = d_t - v_t$ with $\alpha = 1$,*

$$\mathrm{Var}(d_t) = \frac{2a^2}{1+a}\sigma^2, \qquad \mathbb{E}[q_t^2] = \frac{2a^4(a+3)}{(1+a)^3}\sigma^2. \tag{S3.23}$$

Consequently,

$$\frac{\mathbb{E}[q_t^2]}{\mathrm{Var}(d_t)} = \frac{a^2(a+3)}{(1+a)^2}, \tag{S3.24}$$

which is generally not one (it equals approximately 0.875 when $a = 0.9$).

Proof. Let $z_t = m_t - \mu$. Then $z_t = az_{t-1} + (1-a)\xi_t$ and $d_t = \xi_t - z_t = a(\xi_t - z_{t-1})$. Also $v_t = av_{t-1} + (1-a)d_t$. The state vector $x_t = (z_t, v_t)^\top$ follows

$$x_t = \begin{pmatrix} a & 0 \\ -a(1-a) & a \end{pmatrix} x_{t-1} + \begin{pmatrix} 1-a \\ a(1-a) \end{pmatrix} \xi_t.$$

Solving the stationary discrete Lyapunov equation for $P = \mathbb{E}[x_t x_t^\top]/\sigma^2$ gives

$$P_{11} = \frac{1-a}{1+a}, \quad P_{12} = \frac{a(1-a)}{(1+a)^2}, \quad P_{22} = \frac{2a^2(1-a)}{(1+a)^3}.$$

Moreover, (S3.7) gives $q_t = a^2\xi_t - a^2 z_{t-1} - av_{t-1}$. Since ξ_t is independent of the previous state, substituting the entries of P yields the second formula in (S3.23). The variance of $d_t = a(\xi_t - z_{t-1})$ gives the first formula. □

Remark S3.16. *The calculation in Proposition S3.15 is not a model of a full nonstationary training trajectory; it is a counterexample to the general identity $\mathbb{E}[(d_t - v_t)^2] = \mathrm{Var}(d_t)$. It shows that even under a simple stationary oracle, the centering mechanism changes the scale of the accumulated statistic. A safer description is "the EMA of the squared centered prediction-error residual" rather than "an unbiased estimate of the variance of the prediction error."*

S3.7 Role of ϵ and relationship to AdaBelief

For the AdaBelief implementation considered in the manuscript, which adds ϵ at every step of the second-moment recursion,

$$s_t = \beta_2 s_{t-1} + (1-\beta_2) d_t^{\odot 2} + \epsilon \mathbf{1},$$

zero initialization gives the exact bias-corrected decomposition

$$\widehat{s}_t = \frac{(1-\beta_2)\sum_{r=1}^t \beta_2^{t-r} d_r^{\odot 2}}{1-\beta_2^t} + \frac{\epsilon}{1-\beta_2}\mathbf{1}.$$

Hence its denominator has the coordinatewise lower bound $\sqrt{\epsilon/(1-\beta_2)} + \epsilon$. By contrast, EMOJI has $\sqrt{\widehat{j}_{t,i}} + \epsilon \geq \epsilon$ only. Removing the recursive ϵ removes the persistent additive floor $\epsilon/(1-\beta_2)$ from the bias-corrected state, but it also allows the adaptive multiplier to become as large as $1/\epsilon$. Thus post-root ϵ guarantees that division is finite for a fixed positive ϵ, but it does not provide a bound uniform as $\epsilon \downarrow 0$.

This distinction is explicit in the convergence constants: $M = 1/\epsilon$ and $P = G/\epsilon$. For the proposed centered setting with $0 < \beta_1 \leq \beta_2$, $M/\underline{\mu} = (2\gamma G + \epsilon)/\epsilon$; under the generic bound it is $(2G(1+\alpha)+\epsilon)/\epsilon$. The resulting rates are valid for each fixed $\epsilon > 0$, but their constants deteriorate as ϵ decreases. Accordingly, the observed epsilon robustness in the experiments is an empirical property over the tested range, not a uniform-in-ϵ convergence guarantee.

The $\alpha = 0$ variant is an AdaBelief-type update with the recursive ϵ removed and with ϵ retained after the square root. It is not identical to the original AdaBelief implementation used in the manuscript's comparison. The $\alpha = 1$ variant adds the extra state v_t and the filtered residual q_t, so its second-moment dynamics require new analysis.

S3.8 Relationship to existing adaptive-optimizer analyses

The nonconvex stochastic convergence theorem in the original AdaBelief analysis assumes, among other conditions, bounded true and noisy gradients, bounded noise variance, unbiased independent gradient noise, a positive lower bound on the initial second-moment coordinates, and a coordinatewise nondecreasing second-moment sequence s_t [3]. The paper notes that enforcing this monotonicity in the implementation requires an elementwise maximum operation. The EMOJI recursion analyzed here does not impose such a maximum: j_t may decrease and may be zero, while only the final denominator is bounded below by ϵ. Therefore, the original AdaBelief theorem cannot be transferred directly to EMOJI. The deterministic and stochastic guarantees in this supplement are proved for the EMOJI core update under their own stated assumptions.

The broader adaptive-optimization literature also shows that exponential moving-average preconditioners can fail to converge in some settings without suitable temporal control [9]. Analyses of adaptive methods in nonconvex stochastic optimization likewise retain explicit dependence on gradient-noise variance, providing relevant context for the variance term in Theorem S3.12 [11]. FastAdaBelief studies a different algorithm under strong-convexity and algorithm-specific conditions [16]; its guarantees likewise do not follow merely from EMOJI's relation to AdaBelief. These comparisons motivate proving guarantees for the actual recursion rather than relying on results for related optimizers.

S3.9 Scope and limitations of the guarantees

The results above establish the following, and no more:

- For the deterministic core update on an L-smooth, lower-bounded objective with gradients bounded along the trajectory and fixed $\epsilon > 0$, Theorem S3.6 gives a finite-time nonconvex stationarity bound. With a constant step size, the bound is of order $\mathcal{O}((\eta T)^{-1} + \eta)$ when the remaining problem parameters are fixed. Under the polynomially diminishing schedules in Corollary S3.8, the gradient norm tends to zero under the stated assumptions.

- Under the additional PL condition, Corollary S3.9 gives convergence of the deterministic objective gap to zero for the stated diminishing schedules. With a sufficiently small constant step size, it gives geometric contraction to an $\mathcal{O}(\eta)$ objective-gap neighborhood; it does not assert convergence of the parameter vectors themselves.

- For $\alpha = 1$ and $0 < \beta_1 \leq \beta_2$, Proposition S3.1 proves a finite-time reduction of the bias-corrected second-moment state relative to the uncentered accumulator driven by the same prediction-error sequence. This result does not imply pointwise attenuation at every iteration, nor does it compare independent training trajectories.

- For stochastic gradients, Theorem S3.12 gives a finite-time expected-stationarity bound with an explicit filtered-noise term. At fixed batch size and nonzero variance, its limiting time-averaged bound retains a variance-dependent contribution. The expected objective-gap result in Theorem S3.14 requires the additional relative-tracking-error condition stated there.

- No global-optimality guarantee is obtained for a general nonconvex objective, and no convex online-regret guarantee is claimed for the unprojected EMOJI implementation. The constants depend on $1/\epsilon$, so the results are not uniform as $\epsilon \downarrow 0$. The analysis concerns the core adaptive update and does not include the separate decoupled weight-decay step used in the experiments or any gradient-clipping operation. Extending the guarantees to the complete experimental training procedure would require additional terms and assumptions.

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